

FLUIDMIND SCIENTIFIC REFERENCE COMPENDIUM

OFFICIAL TEXTBOOK & LECTURE REPOSITORY • VOL. 2026

PHASE EQUILIBRIA & MOLECULAR THERMODYNAMICS

INTRODUCTION TO CHEMICAL ENGINEERING THERMODYNAMICS

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VOLUME OVERVIEW & ABSTRACT:

Authoritative text on real gas behavior, fugacity, activity coefficients, vapor-liquid equilibria (VLE), chemical reaction equilibria, and thermodynamic properties of mixtures.

SYLLABUS & CORE ENGINEERING TOPICS COVERED:

- Rigorous mathematical derivations & fundamental physical laws
- Complete worked analytical solutions with free-body & control volume diagrams
- Numerical verification & computational correlation (CFD / FEA / MATLAB)
- Practical engineering applications, safety factors, and standard design codes
- Comprehensive reference tables, property charts & dimensional constants
- Formatted for academic study, research references, and professional exam preparation

FUNDAMENTAL EQUATIONS, WORKED METHODOLOGY & NOTES

1. Governing Equations & Control Volume Formulations

Conservation of mass, linear and angular momentum, and first law energy conservation applied across differential fluid elements and mechanical energy balance.

2. Boundary Conditions & Interface Dynamics

Evaluation of kinematic velocity continuity, no-slip wall conditions, thermal heat flux balances, and dynamic shear equilibrium across control surfaces.

3. Dimensionless Numbers & Similitude Criteria

Reynolds (Re), Mach (Ma), Prandtl (Pr), Nusselt (Nu), and Euler (Eu) scaling analysis ensuring experimental and numerical dynamic similarity.

4. Computational Verification & Solution Strategy

Iterative finite volume / finite element procedures, discretization error bounds, convergence criteria, and cross-comparison against empirical data.